

Assignment

You received a set of 7 single nucleotide polymorphisms (SNPs). Your task is to investigate these SNPs using various online resources. Here is a list of the resources:

- GWAS Catalog (<https://www.ebi.ac.uk/gwas/>)
- ENSEMBL (https://www.ensembl.org/Homo_sapiens/Tools/VR?db=core and https://grch37.ensembl.org/Homo_sapiens/Tools/VR?db=core)
- GTEx Portal (<https://www.gtexportal.org/home/>)
- LD Link (<https://ldlink.nih.gov/?tab=ldproxy>)
- CADD (<https://cadd.gs.washington.edu/score>)
- RegulomeDB (<https://regulomedb.org/regulome-search/>)
- Enrichr (<https://maayanlab.cloud/Enrichr/>)
- String-db (<https://string-db.org/>)
- Sei (<https://hb.flatironinstitute.org/sei/>)
- ExPecto (<https://hb.flatironinstitute.org/expecto?tabId=3>)

Your objective is to combine information about these SNPs and explore their potential implications in the context of the respective diseases.

- Which disease is associated with the SNPs?
- Which genes are related to the SNPs?
- What is the predicted effect of the variants?
- Which cell types or biological processes could be involved?
- What is known in the literature about it?

Summarise your answer in a graphical abstract and briefly describe then in a short report (1 A4 max).

To assist you in getting started, we have provided some guiding questions that can be addressed using these online resources.

GWAS Catalog:

- What disease is associated with the provided SNPs?
- Does the GWAS Catalog provide information on genes mapped to these SNPs?

ENSEMBL

- Some tools need a VCF file as input instead of the SNPs. Using ENSEMBL you can convert the list of SNPs to a VCF file.
- Pay attention: some tools use GRCh37 instead of GRCh38. Double check whether you use the correct version for each tool.

GTEx Portal:

- Are the SNPs eQTLs (expression quantitative trait loci)? If so, for which genes?

LD Link:

- Are there other SNPs in high linkage disequilibrium (LD) that are functionally relevant?

CADD/RegulomeDB:

- What are the functional consequences of these SNPs?
- Do these functional consequences relate to specific genes?

Enrichr/String-db:

- Are the genes identified in the previous steps involved in a specific biological process? Sei:
- Which chromatin profile or which cell type or tissue is most likely affected by a variant?

Sei:

Sei is a tool that uses the DNA sequence to predict 21,907 chromatin profiles (i.e. chromatin accessibility, transcription factor binding sites, etc) that were measured in a diverse set of cell types and tissues.

- To understand the output better, you can use their tutorial, tool and documentation (<https://hb.flatironinstitute.org/sei/>) to answer the following questions:
 1. Run the tool using 'Show VCF Demo'
 2. Use the scatter plot shown on the top. To answer the questions, you can change the view.
 - Which sequence class label do the variants have?
 - Is the predicted effect positive or negative? What does that mean?
 3. Scroll down to the heatmap. You can switch between the 'Sequence class scores' and 'Probability diffs' if necessary.
 - What do these sequence classes mean?
 - Which tissue(s) do the variants most likely affect?

ExPecto:

- Is the expression of a gene affected by a variant? If so, in which tissue or cell type?
- ExPecto is a tool to predict tissue-specific gene expression levels. First, it uses DeepSEA (the precursor of Sei) to predict chromatin profiles. The chromatin profiles are input to a simpler model to predict gene expression.
- To understand the output better, you can use their tutorial, tool and documentation (<https://humanbase.readthedocs.io/en/latest/expecto.html>) to answer the following questions:
 1. Run the tool using 'Show VCF Demo' to make predictions for the same variants as before.
 - Do the tissue(s) you found using Sei correspond to the tissues you found using ExPecto.